

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

MODEL QUESTION PAPER – SET-1

Digital Communication

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	List the basic elements of PCM system	M 1.03	R
2	Define Granular noise	M 1.04	R
3	List any two data transmission methods	M 2.03	R
4	Name the modulation scheme which is called as on-off keying	M 2.01	R
5	Define Entropy	M 3.01	R
6	Name any two error detection and correction codes	M 3.04	R
7	Define spread spectrum	M 4.01	R
8	Name any two multiple access techniques	M 4.02	R
9	State Shannon-Hartley theorem	M 3.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

Module Outcome Cognitive level

1	Explain slope overload noise in DM.	M 1.04	U
2	For a PCM system with quantization level 2 have a bandwidth of 1000 samples/sec. Interpret the bandwidth of same PCM system having quantization level 8.	M 1.03	U
3	List the necessity of digital communication.	M 1.01	R
4	Explain the quantization noise in PCM.	M 1.03	U
5	Explain TDMA.	M 4.02	R
6	Describe pseudo-noise sequences.	M 4.01	U
7	Describe OFDM.	M 4.03	U
8	Explain DSSS	M 4.01	U
9	Describe the full duplex data transmission.	M 2.03	U

10	A parity check code has the parity check matrix $[H] = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}.$ Construct the generator matrix.	M 3.04	A
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PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Draw the block diagram of digital communication system and define the functions of each block.	M 1.01	R
OR			
2	Explain Adaptive Delta Modulation with diagram.	M 1.04	U
3	Compare TDM and FDM	M 2.02	U
OR			
4	Explain the concept of QPSK with waveforms.	M 2.01	U
5	An even parity hamming code for 5 data bits is transmitted and received. The received codeword is 110001101 . Detect the error and correct it.	M 3.04	A
OR			
6	A system has a bandwidth of 4kHz and a SNR of 28dB at the input to the receiver. Calculate the information carrying capacity of the system.	M 3.02	U
7	Compare FDMA and CDMA	M 4.02	U
OR			
8	Explain FHSS	M 4.01	U
9	Sketch the state transition diagram for a binary convolution encoder with k=1, n=2 and K =3	M 3.04	A
OR			
10	Name the coding requirements and list the need of coding	M 3.03	R
11	Explain the generation and reception of BFSK	M 2.01	U
OR			
12	Compare synchronous and asynchronous data transmission methods	M 2.03	U